Lecture 4: Computer Architecture & IAS Machine (MCQs)

1. Computer ________ refers to those attributes that have a direct impact on the logical execution of a program.
a) Organization b) Architecture c) Design d) Implementation

• **Answer:** b) Architecture

2. Which of the following is an attribute of Computer Organization?
a) Instruction set b) The number of bits used to represent data types c) Hardware details transparent to the programmer d) I/O mechanisms

• **Answer:** c) Hardware details transparent to the programmer

3. Computer ________ helps in optimizing computer programs performance.
a) Architecture b) Organization c) Assembly d) Logic

• **Answer:** b) Organization

4. The IAS computer memory consists of 1024 words, where each word is ________ bits. a) 20 b) 32 c) 40 d) 64

• **Answer:** c) 40

5. In IAS machine, each instruction consists of an 8-bit opcode and a ________ address. a) 8-bit b) 10-bit c) 12-bit d) 20-bit

• **Answer:** c) 12-bit

6. Which register contains the address of the next instruction pair to be fetched from memory? a) IR b) MAR c) PC d) IBR

• **Answer:** c) PC (Program Counter)

7. Which register is employed to hold temporarily the *right-hand* instruction from a word in memory? a) IR b) MBR c) IBR d) MAR

• **Answer:** c) IBR *(Important: This is a very typical Dr. Shahenda question style)*

8. The register that specifies the address in memory of the word to be written from or read into MBR is ________. a) PC b) MAR c) IR d) AC

• **Answer:** b) MAR

9. In the IAS structure, the _______ holds the results of ALU operations and the most significant bits of a product. a) MQ b) AC c) MBR d) IBR

• **Answer:** b) AC (Accumulator)

10. Which register holds the *quotient* after a division operation in IAS?
a) AC b) MQ c) MBR d) DR

• **Answer:** b) MQ (Multiplier Quotient)

11. The instruction S(x) -> AC performs which operation?
a) Adds content of x to AC
b) Copies number in location x to AC
c) Stores AC into location x
d) Jumps to location x

• **Answer:** b) Copies number in location x to AC

12. The IAS instruction Go to M(x, 0:19) represents a ________.
a) Conditional Branch
b) Unconditional Branch to Left Instruction
c) Unconditional Branch to Right Instruction
d) Data Transfer

• **Answer:** b) Unconditional Branch to Left Instruction

13. For a conditional branch in IAS (e.g., If AC >= 0 ...), the condition checks if the number in the Accumulator is ________.
a) Zero only b) Negative c) Non-negative d) Odd

• **Answer:** c) Non-negative (>= 0)

14. In the Fetch Cycle of IAS, if the instruction is NOT in the IBR, the word is first loaded into ________.
a) IR b) AC c) MBR d) MAR

• **Answer:** c) MBR

15. Which part of the Von Neumann architecture is responsible for controlling the proper sequencing of operations?
a) ALU b) Memory c) I/O d) Control Unit (CU)

• **Answer:** d) Control Unit (CU)

16. The IAS machine stores both ________ and ________ in the same memory.
a) Data, Instructions b) Registers, ALU c) Inputs, Outputs d) Cache, Disk

• **Answer:** a) Data, Instructions

17. AC < AC + M(x) is the symbolic representation for which instruction?
a) SUB M(x) b) ADD M(x) c) MUL M(x) d) STOR M(x)

• **Answer:** b) ADD M(x)

18. The instruction that modifies the address part of another instruction in memory is ________.
a) S(x) -> AC b) M(x, 28:39) -> AC(28:39) c) Ap -> S(x) d) Go to M(x)

• **Answer:** c) Ap -> S(x) (Replace address part),

19. In IAS, the MUL instruction places the *least significant bits* of the result in ________.

a) AC b) MQ c) MBR d) MAR

• **Answer:** b) MQ

20. Which of the following is TRUE about Computer Architecture?

a) It is decided after the organization is fixed.
b) It describes "how" the computer does it.
c) It is visible to the programmer.
d) It changes frequently with every new model.

• **Answer:** c) It is visible to the programmer

21. The "Stored Program Concept" is attributed to _________.

a) Charles Babbage b) Von Neumann c) Alan Turing d) Moore

• **Answer:** b) Von Neumann

22. In the instruction L (Shift Left) of IAS, the number in AC is shifted left by 1 bit, effectively _________.

a) Multiplying by 2 b) Dividing by 2 c) Adding 1 d) Doing nothing

• **Answer:** a) Multiplying by 2,

23. The JUMP M(X, 20:39) instruction transfers control to the _________ instruction at location X.

a) Left-hand b) Right-hand c) Middle d) Next

• **Answer:** b) Right-hand

24. Which register contains the 8-bit op-code instruction currently being executed?

a) MBR b) PC c) IR d) IBR

• **Answer:** c) IR

25. When the processor fetches an instruction pair from memory to MBR, the *left* instruction goes to IR/MAR and the *right* instruction is sent to _________.

a) PC b) IBR c) AC d) MQ

• **Answer:** b) IBR

26. The R (Shift Right) instruction in IAS shifts the number in AC to the right by 1 bit, effectively _________.

a) Multiplying by 2 b) Dividing by 2 c) Subtracting 1 d) Inverting sign

• **Answer:** b) Dividing by 2,

27. STOR M(X) is an example of a _________ instruction.
a) Control b) Arithmetic c) Data Transfer d) Branch

• **Answer:** c) Data Transfer (Move)

28. An IAS instruction pair consists of two instructions packed into a single word. Instruction 1 occupies bits _________.
a) 0-19 b) 20-39 c) 0-8 d) 28-39

• **Answer:** a) 0-19

29. To execute a program, the Control Unit performs an instruction cycle consisting of _________.
a) Fetch and Decode b) Fetch and Execute c) Read and Write d) Load and Store

• **Answer:** b) Fetch and Execute

30. Which IAS instruction clears the AC and subtracts the absolute value of the number at memory location X?
a) S(x) -> Ac-M b) S(x) -> Ac- c) S(x) -> Ah-M d) S(x) -> AhM

• **Answer:** a) S(x) -> Ac-M (Load Negative Absolute)